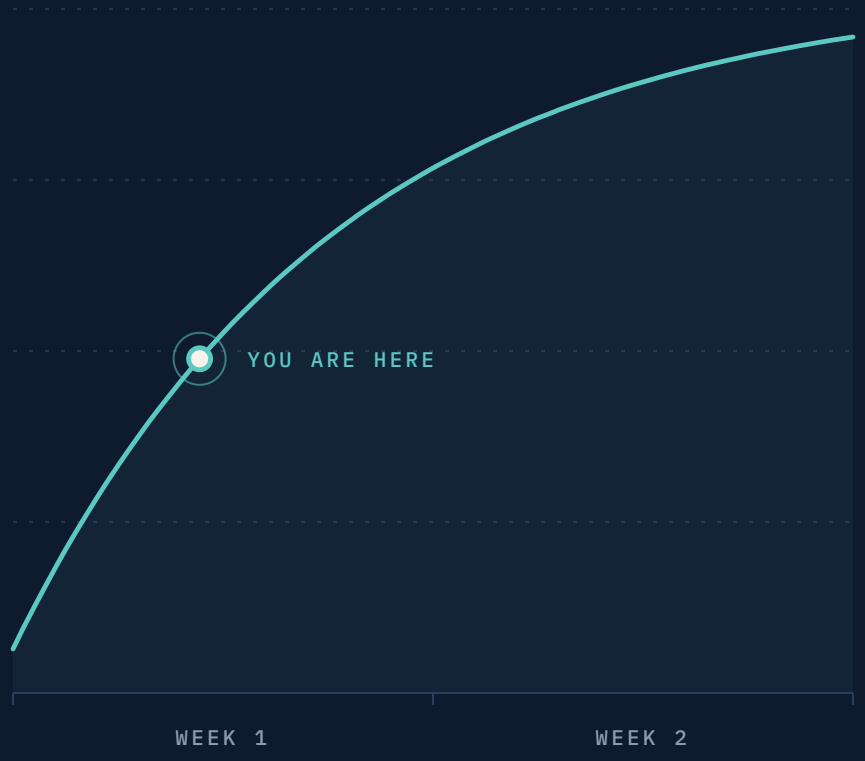
WEEK 1 · DAY 3 · OVERVIEW

Stationarity and Differencing



Created by **Jeff Lenamon**

ABOUT 50 MIN WITH EXERCISES / 11 SECTIONS / 5 EXERCISES

This course is educational content created in a personal capacity. Nothing here is investment advice or a recommendation to buy or sell any security. All portfolio data is synthetic.

Problem Statement

BUSINESS CONTEXT

Day 2 set the bar every model must beat, and one rule fell far behind it: the training mean missed the 10 Yr's level by many times what the last value carried forward missed.

Days 4 to 9 fit models, and each must be told which series to model, the level or the daily change.

A colleague also hands over a regression of the 10 Yr's level on the IG spread's level with a slope p-value far below 0.05, yet the spread file was generated with no reference to yields.

Today finds which series a model may use, with two tests that read in opposite directions, and shows why that regression misleads.

OBJECTIVES

Draw a trailing mean and spread, simulate series from one set of shocks and see what one setting does to them, and run the ADF and KPSS tests, each against its own null hypothesis.

Read the two tests together on five series, regress one wandering series on another, and difference a series once, twice, and after a log.

ABOUT 50 MIN WITH EXERCISES / 11 SECTIONS / 5 EXERCISES

Your Route Through Day 3

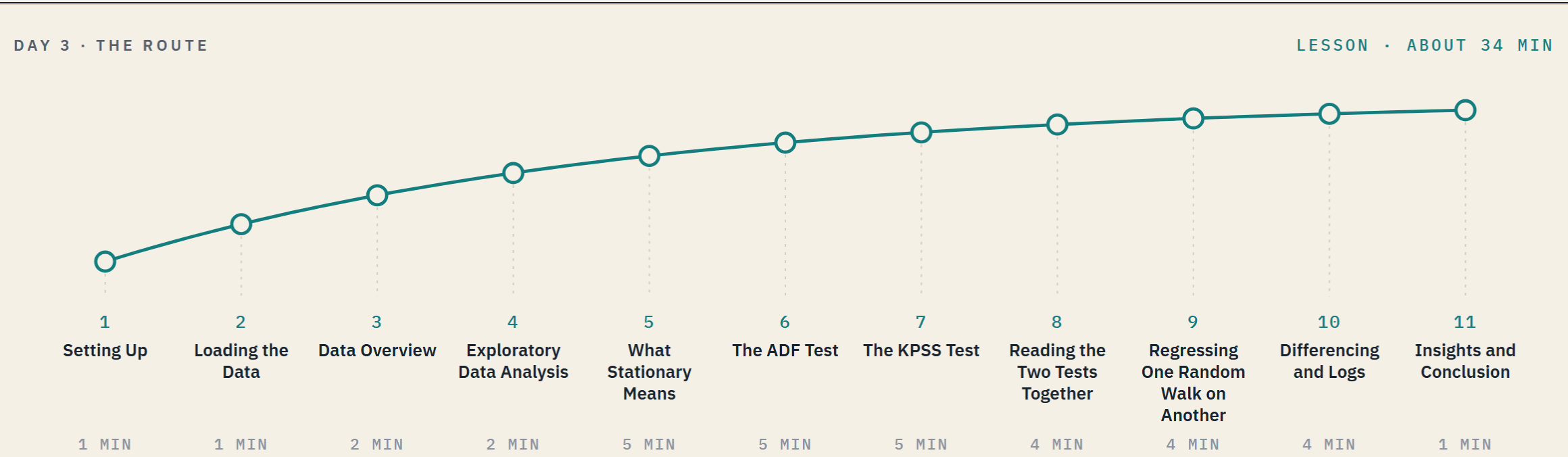


FIGURE 1 Eleven sections, in order, with about how long each takes.

Time. About 34 minutes for the lesson, then about 15 for the five exercises.

Fix the seed and load the libraries.

Setting Up

THE SETUP

```
import os
from pathlib import Path
os.environ["OMP_NUM_THREADS"] = "1"
os.environ["OPENBLAS_NUM_THREADS"] = "1"
SEED = 42
DATA = "../../../data/"
if not Path(DATA + "benchmark.csv").exists():
    DATA = "https://raw.githubusercontent.com/lenamonj/python-fixed-income/main/data/"
```

THE LIBRARIES

```
import numpy as np
import pandas as pd
from statsmodels.tsa.stattools import adfuller
from arch.unitroot import KPSS
```

Read both files and build five series.

Loading the Data

READING THE TWO FILES

```
yields = pd.read_csv(DATA + "treasury_par_yields.csv", parse_dates=["date"], index_col="date")
spreads = pd.read_csv(DATA + "course3_synthetic_spreads.csv", parse_dates=["date"], index_col="date")
CUT = "2022-12-30"
development = yields.loc[:CUT]
```

FIVE SERIES IN BP

```
levels = pd.DataFrame({"2 Yr": (development["2 Yr"] * 100).round(2),
                       "10 Yr": (development["10 Yr"] * 100).round(2),
                       "2s10s": ((development["10 Yr"] - development["2 Yr"]) * 100).round(2),
                       "IG": spreads.loc[:CUT, "ig_spread_bp"],
                       "HY": spreads.loc[:CUT, "hy_spread_bp"]})
changes = levels.diff().dropna()
print(levels.shape, changes.shape)
```

```
(2001, 5) (2000, 5)
```

Data Overview

THE LEVELS, IN BP

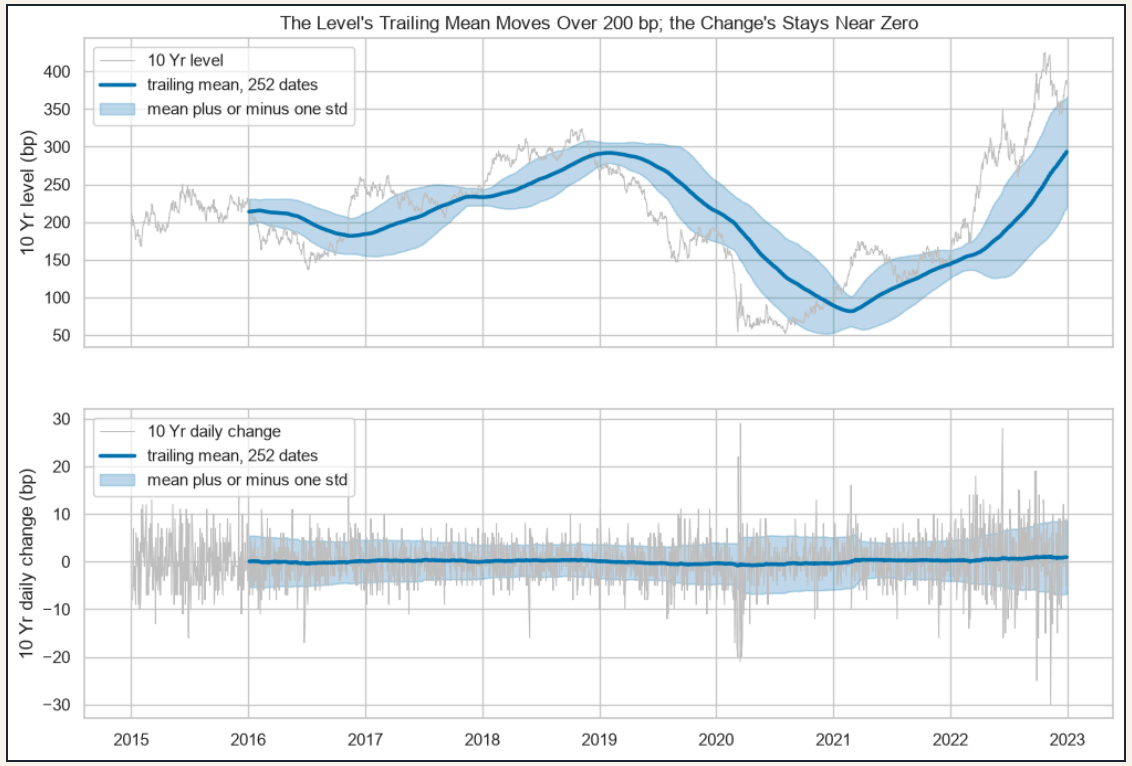
	2 YR	10 YR	2S10S
count	2001.0000	2001.0000	2001.0000
mean	137.9265	207.8826	69.9560
std	106.0845	73.8736	53.4732
min	9.0000	52.0000	-84.0000
25%	60.0000	160.0000	26.0000

THE DAILY CHANGES, IN BP

	2 YR	10 YR	2S10S
count	2000.0000	2000.0000	2000.0000
mean	0.1875	0.0880	-0.0995
std	4.1905	5.0285	3.3230
min	-27.0000	-30.0000	-16.0000
25%	-2.0000	-3.0000	-2.0000

Five complete series. The levels roam over hundreds of bp while the changes sit around zero, so the next question is whether each keeps one average over the eight years.

Exploratory Data Analysis



HOW FAR EACH TRAILING MEAN MOVED

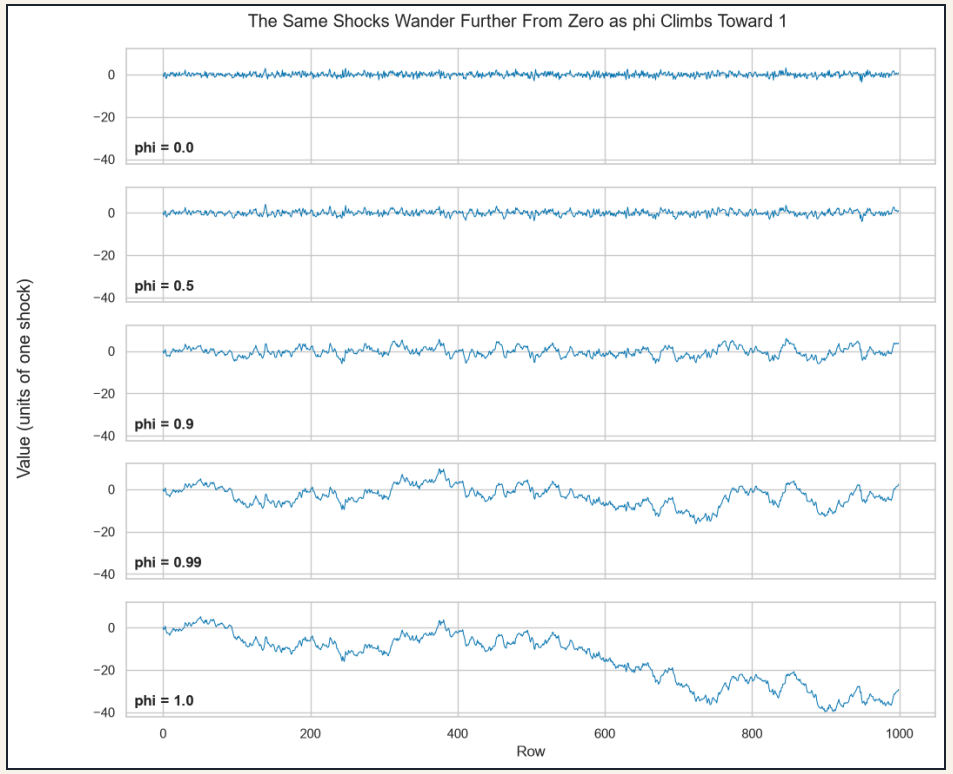
```
level: trailing mean from 81.60 to 293.37 bp, a move of 211.77
change: trailing mean from -0.85 to 1.06 bp, a move of 1.91
change: trailing std from 3.23 bp (2018-05-21) to 7.75 bp (2022-12-30)
```

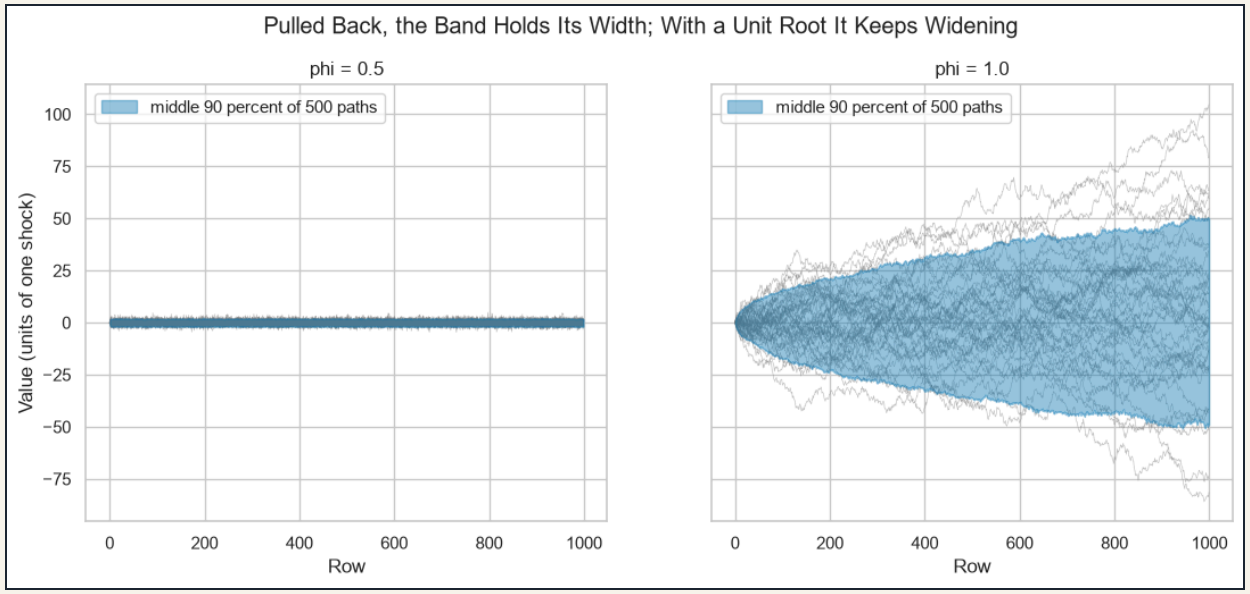
THE SPREAD

The change's spread did not hold still: from 3.23 bp in May 2018 to 7.75 at the end of 2022.

No single average describes the level, so Day 2's training mean had nothing to hold on to. The change keeps its average. The word for that is stationary.

What Stationary Means





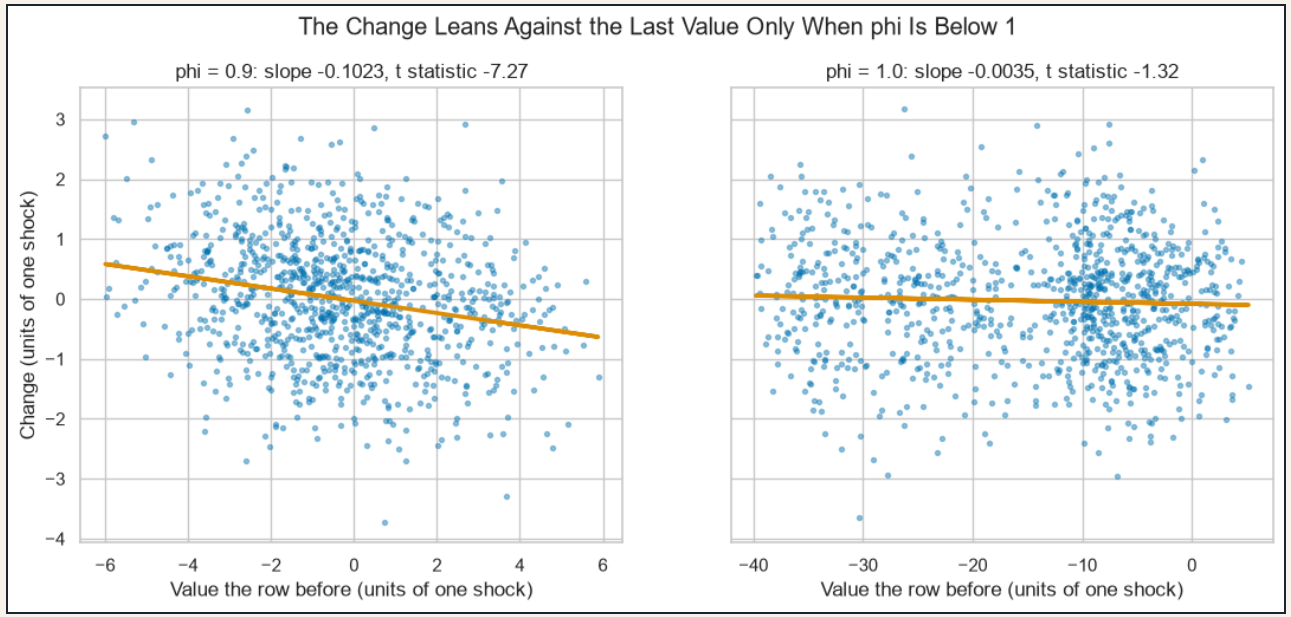
THE BAND'S WIDTH AT THREE ROWS

```
phi = 0.5: width at row 50 3.9, row 250 3.7, row 999 3.5
phi = 1.0: width at row 50 23.1, row 250 50.0, row 999 99.0
```

A unit root shows as a spread that grows with time. We need a test that sees it in one series.

Run ADF and see the regression beneath.

The ADF Test



ADF ON THE 10 YR

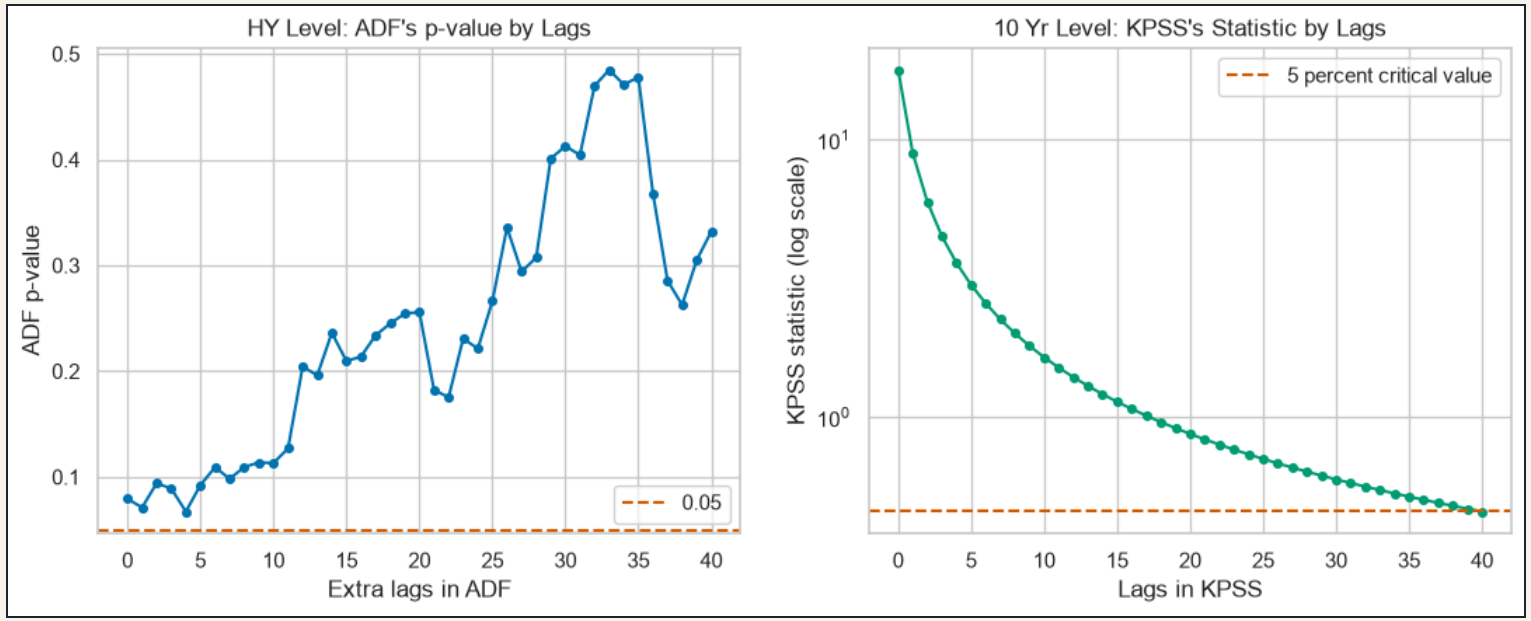
```
level:  p-value 0.87, lags 10
change: p-value 2.09e-24, lags 9
```

THE REGRESSION BY HAND

```
by hand:  slope -0.00083, t statistic -0.5456
adfuller: statistic -0.5456, 5 percent critical value -2.86
```

ADF is Course 4's regression read against its own table. It asks whether the last level pulls the change back.

The KPSS Test



WHAT KPSS DOES

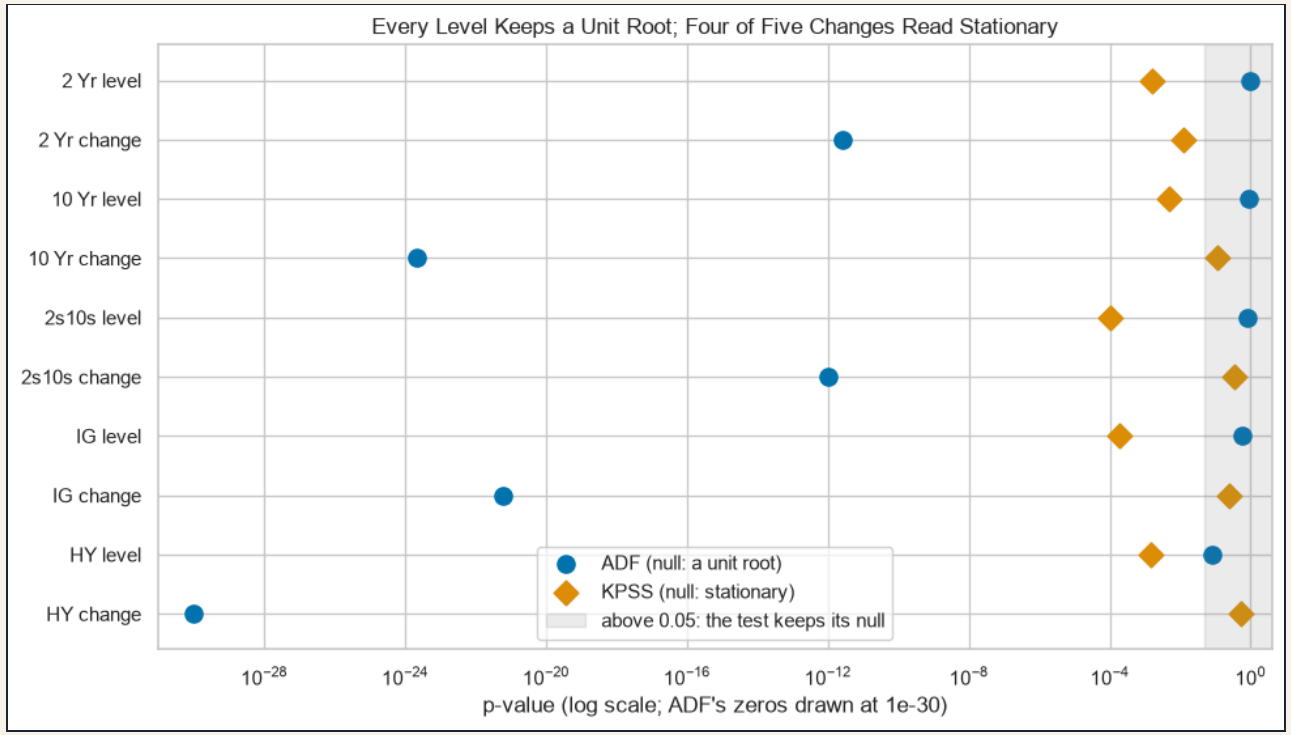
It takes the series' own average away, keeps a **running total** of what is left, and rejects when that total swings far from zero.

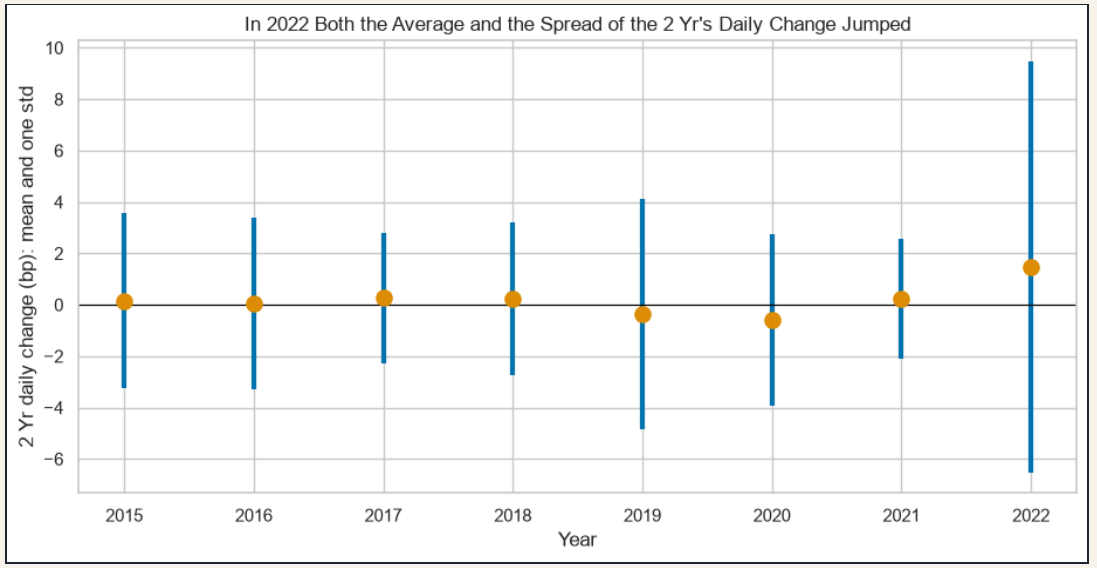
THE QUESTION

Q. Sketch the running total of the `phi` = 1 walk's distance from its own average, row 0 to row 999.

Settings are chosen before the answer is seen: this course keeps ADF's `autolag="AIC"` and KPSS at `lags=20`.

Reading the Two Tests Together



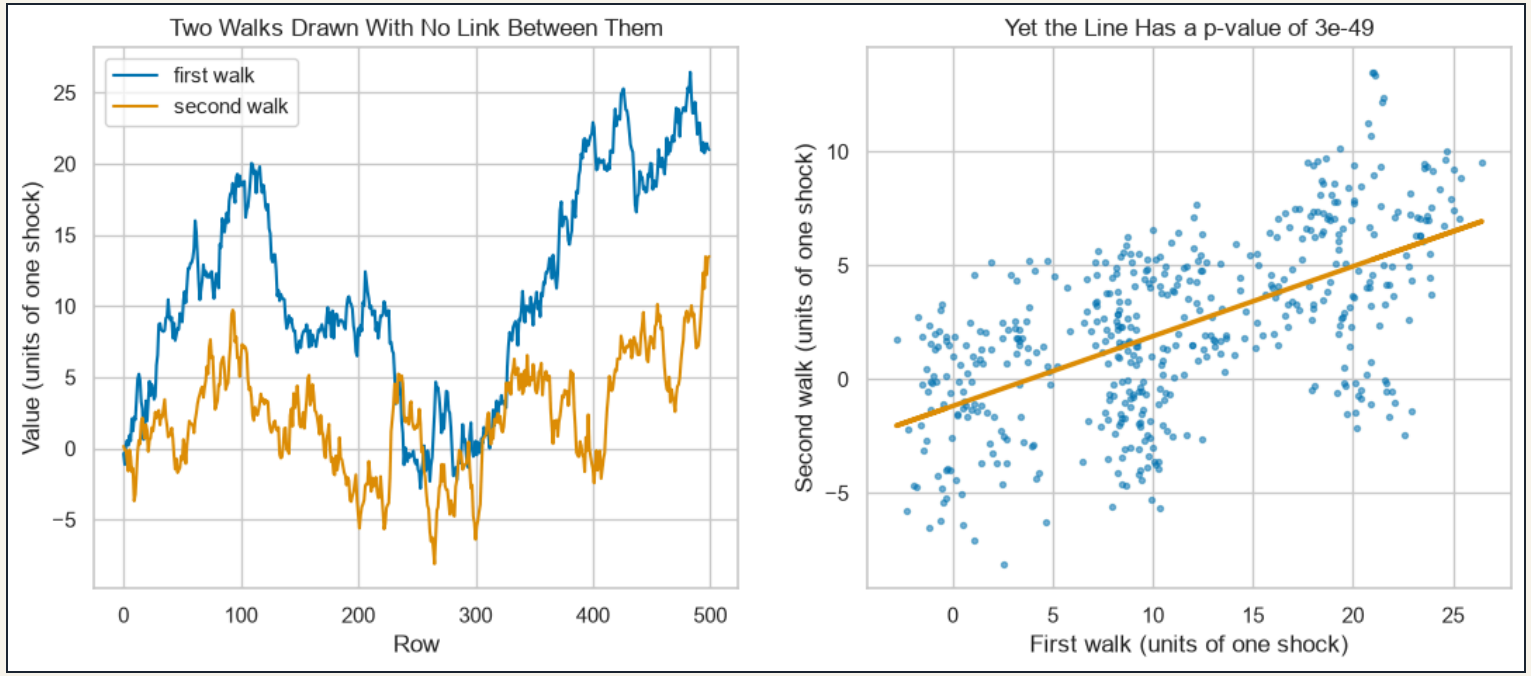


THE ONE THAT FAILS

The 2 Yr's change: both reject.

One difference removes a unit root, not a shift in the change's average or spread. Days 4 to 9 work on changes, and Day 7 takes up the spread.

Regressing One Random Walk on Another



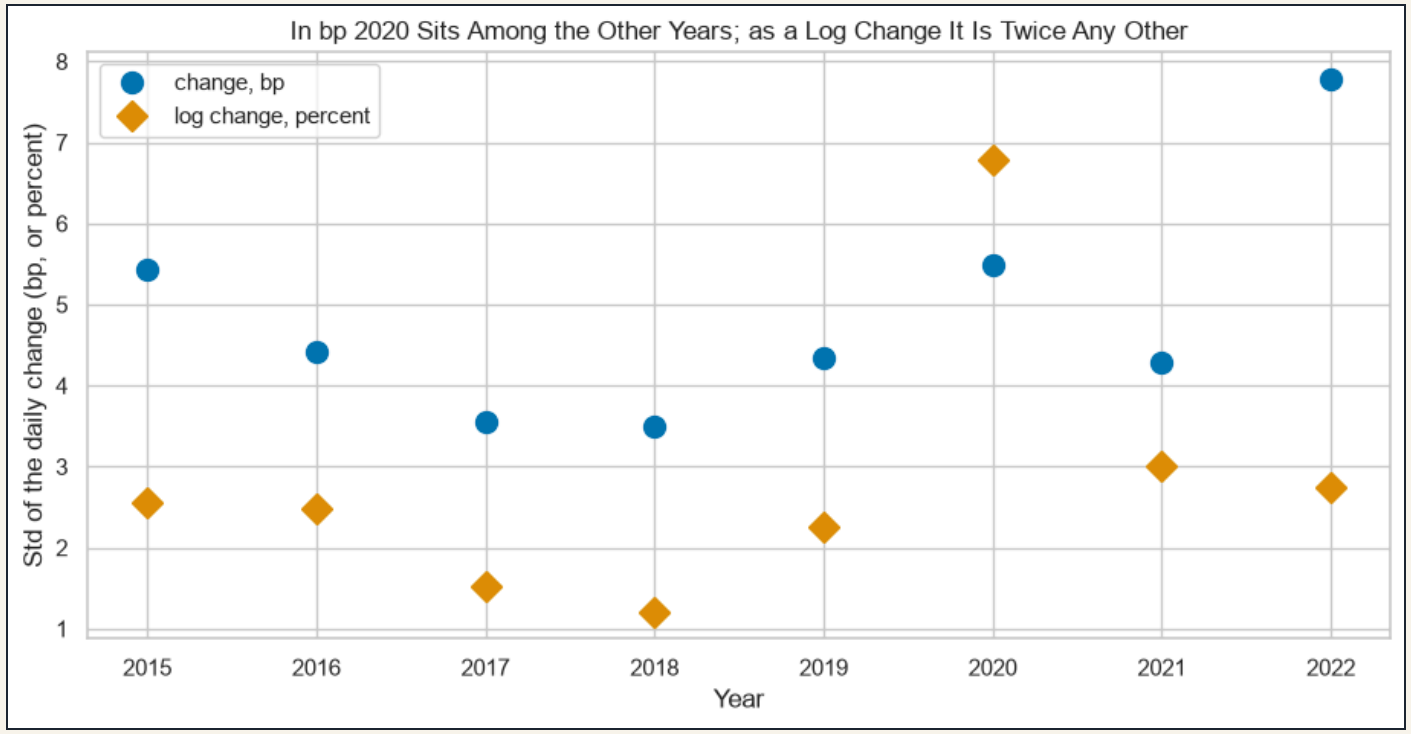
THE QUESTION

Q. Over 1,000 pairs of independent walks, what share of slopes will have a p-value under 0.05? Write down a number.

ONE PAIR OF INDEPENDENT WALKS

```
slope 0.307, p-value 2.8e-49, R-squared 0.354, Durbin-Watson 0.109
```

Differencing and Logs



A LOG CHANGE

It suits a series that compounds, such as a price index; we make one from the generator and compare.

THE QUESTION

Q. What lag-1 autocorrelation will its second difference have? Write down a number.

Logs suit a series that compounds. Yields and spreads in bp are differenced, not logged.

Insights and Conclusion

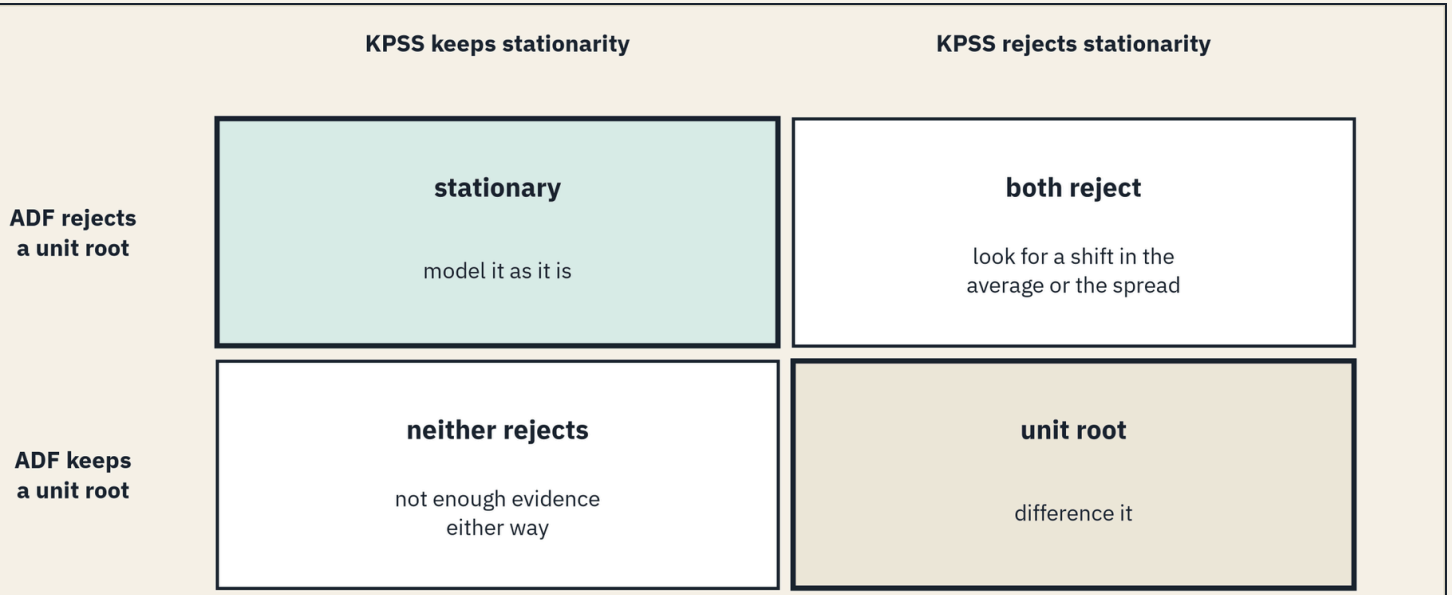


FIGURE 3 The four readings of ADF and KPSS together: stationary, unit root, both reject, and neither rejects.

TWO TESTS, FOUR READINGS

Two tests with opposite null hypotheses give four readings.

Every later day models changes in bp, tested first.

You Can Now

- Draw a trailing mean with a band and say whether a series keeps one average.
- Build an AR(1) from shocks and say what **`phi`** does to it.
- Run ADF with **`adfuller`** and KPSS with arch's **`KPSS`**, and read each against its null hypothesis.
- Read the two tests together in one of four readings.
- Spot a regression between two wandering series, and difference a series the right number of times.

NEXT

WEEK 1 · DAY 4

Autocorrelation: ACF and PACF

Day 4 looks at the links from one day to the next, with the ACF and PACF.

